

Uni-Perceiver v2: A Generalist Model for Large-Scale Vision and Vision-Language Tasks

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Abstract

Despite the remarkable success of foundation models, their task-specific fine-tuning paradigm makes them inconsistent with the goal of general perception modeling. The key to eliminating this inconsistency is to use generalist models for general task modeling. However, existing attempts at generalist models are inadequate in both versatility and performance. In this paper, we propose Uni-Perceiver v2, which is the first generalist model capable of handling major large-scale vision and vision-language tasks with competitive performance. Specifically, images are encoded as general region proposals, while texts are encoded via a Transformer-based language model. The encoded representations are transformed by a task-agnostic decoder. Different tasks are formulated as a unified maximum likelihood estimation problem. We further propose an effective optimization technique named Task-Balanced Gradient Normalization to ensure stable multi-task learning with an unmixed sampling strategy, which is helpful for tasks requiring large batch-size training. After being jointly trained on various tasks, Uni-Perceiver v2 is capable of directly handling downstream tasks without any task-specific adaptation. Results show that Uni-Perceiver v2 outperforms all existing generalist models in both versatility and performance. Meanwhile, compared with the commonly-recognized strong baselines that require tasks-specific fine-tuning, Uni-Perceiver v2 achieves competitive performance on a broad range of vision and vision-language tasks.

1. Introduction

Learning a general perception model that can handle various modalities and tasks is widely regarded as an im-

portant step towards artificial general intelligence. Due to its difficulty, many works (e.g., Florence [45], CoCa [44], BEiT-3 [40]), also known as *foundation models* [2], instead focus on a fallback solution of learning a general representation encoder that can be adapted (e.g., fine-tuned) to various downstream tasks. By performing large-scale pre-training on massive multi-modal task-agnostic data, these works have demonstrated the superiority by pushing the state-of-the-art results on a broad range of tasks including single-modal tasks (e.g., image classification and object detection) and also cross-modal tasks (e.g., image captioning and image retrieval).

Despite the success, there is still a considerable gap between foundation models and the goal of general perception modeling. While foundation models only focus on general representation learning, task modeling is neglected. Traditional task-specific fine-tuning paradigm is still utilized (see Fig. 1). This significantly increases the marginal cost of adapting pre-trained models to various downstream tasks, making it difficult to meet the rapidly growing demands of diverse downstream tasks and scenarios. Such a task-specific fine-tuning paradigm of foundation models is inconsistent with the goal of general perception modeling.

Instead of performing task-specific fine-tuning, generalist models process different tasks with shared architecture and parameters, which is aligned with the goal of general perception modeling. It not only reduces the cost of handling diverse tasks but also enables task collaboration. Most existing attempts on generalist models are sequence-to-sequence (seq2seq) models [1, 5, 10, 14, 23, 29, 39, 43]. However, these attempts are inadequate in both versatility and performance: (1) some pillar vision and vision-language tasks as listed in Tab. 1 cannot be handled, e.g., image-text retrieval, object detection, and instance segmentation; (2) the accuracy and inference speed still lag significantly behind state-of-the-art task-specific methods. Another line of research named Uni-Perceivers [49, 50] builds generalist models supporting both generation and

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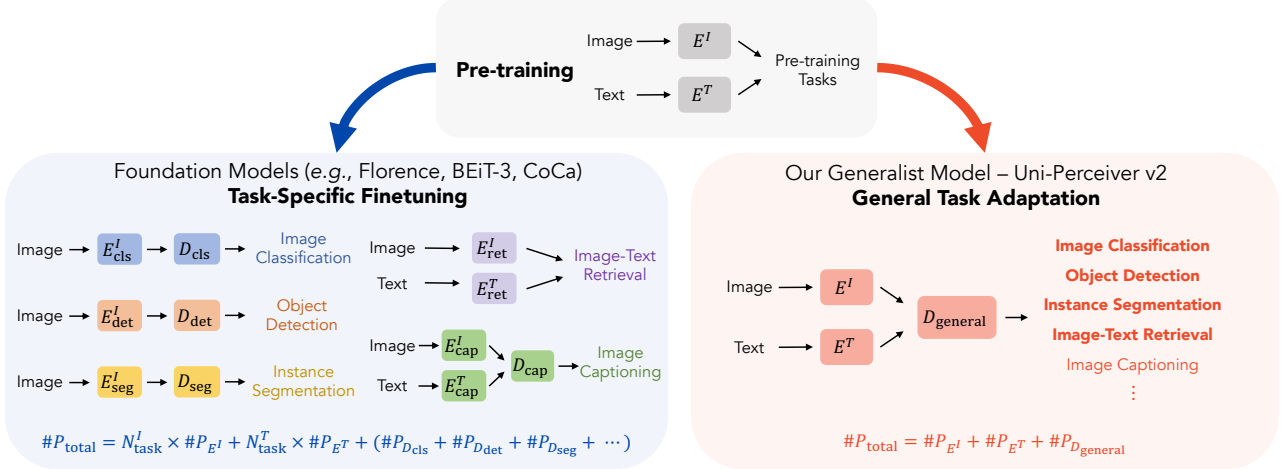


Figure 1. Comparison of foundation models and Uni-Perceiver v2. E^I and E^T denote the image encoder and text encoder, respectively. In existing foundation models, task-specific decoders D_{cls} , D_{det} , $\dots$ are employed to tune E^I and E^T in different task-specific finetuning. The total number of parameters $\#P_{total}$ in adaptation grow with the number of visual/linguistic tasks, denoted as N_{task}^I and N_{task}^T , respectively. By contrast, our Uni-Perceiver v2 shares all parameters across various downstream tasks with a general decoder $D_{general}$, where no task-specific fine-tuning is incorporated. Better than previous generalist models, our method can also effectively handle pillar tasks such as image classification, object detection, instance segmentation, and image-text retrieval.

non-generation tasks. Nevertheless, they still cannot handle many vital tasks such as detection and segmentation.

To develop generalist models with better versatility and performance, our core idea is to encode images as general region proposals consisting of the semantic, bounding box and segmentation mask representations. Compared with previous methods where images are represented as non-overlapping patches, this design makes our localization modeling more expressive and flexible. This explicit encoding of foreground information not only greatly reduces the difficulty of handling localization tasks such as image detection and segmentation, but also provides richer features for understanding textual concepts in non-localization vision-language tasks, thus enabling more general task modeling and better performance.

In this paper, we propose Uni-Perceiver v2 as a generalist model capable of handling major large-scale vision and vision-language tasks as listed in Tab. 1. Specifically, images are encoded as a concatenation of global and regional representations via a region proposal network, while texts are encoded via a Transformer-based language model. Both the image and text encoders can benefit from off-the-shelf pre-trained models, which reduces the demand for training data and resources and ensures performance. The encoded representations are transformed by a shared modality-agnostic Transformer [36] network to obtain the decoded representations. Following Uni-Perceivers [49, 50], different tasks are formulated as a unified maximum likelihood estimation problem and are jointly learned to enable general task adaptation. We further propose *Task-Balanced Gradient Normalization* to ensure sta-

ble multi-task learning with an unmixed sampling strategy which only samples one task for all GPUs per iteration. This is very helpful for tasks requiring large batch size training.

Uni-Perceiver v2 is the first generalist model achieving competitive results on major large-scale vision and vision-language tasks including object detection, instance segmentation, image classification, image captioning, and image-text retrieval, except for image generation that has not been verified due to limited computational resources. After being jointly trained on various tasks, it can directly handle a broad range of tasks without any task-specific adaption, achieving state-of-the-art performance among existing generalist models. Our contributions are summarized as:

- We propose Uni-Perceiver v2, which is the first generalist model capable of handling both localization and non-localization tasks with competitive performance. The general region proposal encoding of images brings more flexible and expressive localization modeling.
- To improve the effectiveness of multi-task learning, we adopt an unmixed sampling strategy to enable large batch-size training and develop an effective optimization technique named Task-Balanced Gradient Normalization to mitigate the instability in gradients.
- Uni-Perceiver v2 outperforms all existing generalist models in both versatility and performance. Without any task-specific adaption, Uni-Perceiver v2 achieves competitive performance on a broad range of downstream tasks compared with commonly-recognized strong baselines that require task-specific fine-tuning, demonstrating its strong ability of general task modeling.

Categories	Specific Tasks
Retrieval	Image-text retrieval
Classification	Image classification Region categorization Situation recognition
Localization	Object detection Key point detection Pose estimation Referring expression grounding Human object interaction Relation detection Optical character recognition Object localization
Mask Predication	Instance segmentation Semantic segmentation Panoptic segmentation
Image Generation	Image synthesis Image inpainting Segment-based image generation Style transferring Depth estimation Surface normal estimation Image infilling Image super resolution
Image to Text	Image captioning Visual question answering Region captioning Grounded VQA Grounded captioning Visual commonsense reasoning

Table 1. Categories of mainstream vision and vision-language tasks. Pillar tasks of different downstream task categories are in **bold**. These pillar tasks are the most representative tasks in each category, where other tasks can be derived from them. UniPerceiver v2 is able to effectively handle the underlined pillar tasks, except for image synthesis that has not been verified due to limited computational resources.

2. Related Work

Foundation Vision Models are “designed to be adapted (e.g., fine-tuned) to various downstream tasks by *pre-training* on broad data at scale” [2]. Such large-scale pre-trained vision models have shown effectiveness in enriching data encoding capacity, alleviating data hunger, and improving the performance of downstream tasks.

Image classification on ImageNet-1k [8] has been the mainstream pre-training paradigm for a long period. However, as the model size grows, larger annotated datasets are required to avoid over-fitting in pre-training, such as ImageNet-21k [8], Instagram-1B [24], JFT-300M [35] and JFT-3B [46]. Inspired by the success of linguistic pre-training on massive web-crawled text, CLIP [27] and ALIGN [12] have begun to focus on multi-modal contrastive pre-training on web-scale noisy image-text pairs to

learn aligned image and text representations. SimVLM [41] employs the multi-modal sequence generation task for pre-training. FLAVA [34] combines contrastive and generative pre-training to handle both unimodal and multimodal tasks. UniCL [42] and CoCa [44] jointly use human-annotated and web-crawled data. Florence [45] and INTERN [31] increase the scale and diversity of pre-training data to enhance the representation capability. OmniVL [37] proposes to incorporate both image-language and video-language tasks in its pre-training. GLIP [17] and GLIPv2 [48] propose a unified pre-training framework for localization tasks and vision-language understanding tasks. BEiT-3 [40] unifies pre-training objectives for different modalities as a single masked data modeling task, achieving state-of-the-art results on a wide range of downstream tasks.

These works on foundation models only focus on general representation learning, while neglecting task modeling. When adapting them to downstream tasks, the traditional task-specific fine-tuning paradigm is still utilized, which is inconsistent with the goal of general perception modeling. Meanwhile, with the rapidly growing demands of diverse tasks and scenarios, the task-specific fine-tuning paradigm would result in a prohibitive marginal cost for data collection, data annotation, model training, and model storage.

Generalist models handle various tasks with shared architecture and parameters, which have been long pursued by the machine learning community. Recently, inspired by the success of sequence-to-sequence (seq2seq) models in NLP field [28], OFA [39], Flamingo [1], and GIT [38] propose to model various tasks as a sequence generation task. Unified-IO [23], Pix2Seq v2 [5], and UniTab [43] further develop this method to support more tasks by introducing discrete coordinate tokens, thus location information can be encoded or decoded by the unified models. Beyond that, Gato [29] succeeds in unifying reinforcement learning tasks into the seq2seq framework. GPV [10] also builds a general-purpose vision system by adding a seq2seq module on a DETR [3]-based visual encoder.

However, these methods with seq2seq formulation are still inadequate in both versatility and performance: (1) They cannot handle some core vision tasks, e.g., image-text retrieval, object detection, and instance segmentation. Although Pix2Seq v2 [5] includes detection and instance segmentation tasks, its performance and inference speed still lag significantly behind state-of-the-art task-specific methods [16, 47]; (2) The non-parallel auto-regressive decoding leads to slow inference speed. For example, image classification requires calculating and comparing the cumulative probabilities of all category names conditioned on the given image; (3) They also suffer from the task-interference issue in multi-task learning, resulting in performance degradation compared with task-specific models.

Alternatively, Uni-Perceivers [49, 50] formulate different tasks as finding the maximum likelihood target for each input through the representation similarity regardless of their modality, making it possible to support both generation and non-generation tasks. Nevertheless, they still cannot handle image detection and segmentation tasks.

3. Revisiting Uni-Perceivers

Unified Modeling of Perception Tasks. Uni-Perceiver [50] proposes to reformulate different tasks as a unified maximum likelihood estimation problem. Specifically, each task is defined with a set of inputs and a set of candidate targets from arbitrary combinations of modalities. The inputs and targets are first encoded with a modality-specific tokenizer with linear projection. Then the encoded representations are transformed by modality-agnostic decoder with shared parameters for different tasks. Given an input, the unified task objective is defined as finding the target with the maximum likelihood with the input.

Mitigating Task Interference. Multi-task learning with fully shared parameters could introduce interference between different tasks. Uni-Perceiver-MoE [49] proposes Conditional MoEs to address the task-interference issue. Specifically, for each input token, a routing decision is calculated depending on specific routing strategy, which sparsely activates a small portion of experts to process this token. The corresponding output of an input token is the linearly weighted combination of those selected experts by the routing decision. Conditional MoEs mitigate the interference issue by allowing conflicting modalities and tasks using separate parameters without introducing any task-specific modules.

Limitations. Although Uni-Perceivers aim to process different tasks with a unified architecture, it fails to handle detection and segmentation tasks due to the lack of localization information in its encoded features. Meanwhile, Uni-Perceivers do not integrate off-the-shelf encoder models, making it unable to benefit from existing large-scale pre-trained encoders. This potentially increases its demand for pre-training data and resources, limiting its performance.

4. Method

4.1. Encoding Images as General Region Proposals

Most existing generalist models [49, 50] represent images as non-overlapping patches with fixed sizes. This design is rather coarse and limited in modeling objects of varying sizes and shapes in images, making it difficult to handle localization tasks such as detection and segmentation.

In order to enable more expressive and flexible localization modeling, we propose to encode the input image as a sequence of general region proposals. Specifically, given

an input image $x \in \mathbb{R}^{H \times W}$ with height H and width W , a network $f_{\text{image}}(\cdot)$ is employed to encode the image as the concatenation of global and regional representations as

$$f_{\text{image}}(x) = \text{Concat} \left(\{q_i^{\text{global}}\}_{i=1}^M, \{q_j^{\text{proposal}}\}_{j=1}^N \right), \quad (1)$$

where $q_i^{\text{global}} \in \mathbb{R}^d$ are the global representations of the whole image, and $q_j^{\text{proposal}} \in \mathbb{R}^d$ are the regional representations of candidate object proposals in the image. The regional representations embed information of foreground objects, and the global representations complement the regional representations with background scene information.

Following the common practice in localization tasks, an image backbone network (e.g., ResNet [11]) is firstly employed to extract the multi-scale feature maps $\{\mathcal{F}_l\}_{l=1}^L$, where L is the number of feature scales (e.g., $L = 4$).

Regional Representations. A Transformer [36]-based region proposal network is applied on top of the multi-scale feature maps $\{\mathcal{F}_l\}_{l=1}^L$ to extract a set of O candidate object proposals $\{q_j^{\text{sem}}, q_j^{\text{box}}, q_j^{\text{mask}}\}_{j=1}^O$, where $q_j^{\text{sem}} \in \mathbb{R}^d$, $q_j^{\text{box}} \in \mathbb{R}^4$, and $q_j^{\text{mask}} \in \mathbb{R}^{H \times W}$ are the semantic, bounding box, and segmentation mask representations of the j -th proposal, respectively. The region proposal network is similar to MaskDINO [16], but only considers foreground-background binary classification. See Appendix A for detailed implementation. These three representations are then fused as the regional representation as

$$q_j^{\text{proposal}} = q_j^{\text{sem}} + \mathcal{B}(q_j^{\text{box}}) + \mathcal{M}(q_j^{\text{mask}}), \quad (2)$$

where $\mathcal{B}$ denotes the positional encoding of box coordinates. $\mathcal{M}$ uses adaptive average pooling to scale the mask predictions to the size of 28×28 . Both $\mathcal{B}$ and $\mathcal{M}$ are followed by linear projections to match the feature dimension.

Global Representations. The global representations are extracted from the last-scale feature map $\mathcal{F}_L \in \mathbb{R}^{h \times w}$ with height h and width w . M' instances of parameterized Attention Pooling [27] are employed to extract global features. The pooled features are concatenated with the flattened feature map to obtain the global representations as

$$q^{\text{global}} = \text{Concat} \left(\{ \text{AttnPool}_i(\mathcal{F}_L) \}_{i=1}^{M'}, \text{Flatten}(\mathcal{F}_L) \right). \quad (3)$$

4.2. Encoding Text with Language Models

A Transformer [36]-based language model is used to encode textual data, such as category names in classification tasks, image descriptions in image-text retrieval tasks, and the vocabulary in image captioning tasks. Specifically, a BPE tokenizer [30] tokenizes the input text x into a sequence of word embeddings, and a Transformer encoder is employed to extract the text feature sequence as

$$f_{\text{text}}(x) = \text{Concat}(q_1^{\text{text}}, q_2^{\text{text}}, \dots, q_L^{\text{text}}) \quad (4)$$

where $q_i^{\text{text}} \in \mathbb{R}^d$ is the encoded feature of the i -th word, and L is the sequence length. In our implementation, we use a pre-trained RoBERTa_{BASE} [20] as the text encoder, which is jointly tuned with the whole network.

4.3. General Task Adaptation

We follow Uni-Perceivers [49, 50] to formulate different tasks as a unified maximum likelihood estimation problem. Given an input $x \in \mathcal{X}$ and the candidate target set $\mathcal{Y}$, the task objective is defined as finding the target $\hat{y} \in \mathcal{Y}$ with the maximum likelihood as

$$\hat{y} = \arg \max_{y \in \mathcal{Y}} P(x, y), \quad (5)$$

where the likelihood $P(x, y)$ is estimated from the cosine similarity between the representations of x and y as

$$P(x, y) \propto \exp \left(\cos \left(g \circ f(x), g \circ f(y) \right) / \tau \right), \quad (6)$$

where $f(\cdot)$ is the modality-specific encoders f_{image} and f_{text} introduced in Sec. 4.1 and 4.2, respectively. $g(\cdot)$ is a modality-agnostic Transformer [36] network shared for different tasks, and $\tau > 0$ is a learnable temperature parameter.

Depending on task requirements, the modality-specific encoded representation for inputs x can be an image feature sequence $f_{\text{image}}(x)$, a text feature sequence $f_{\text{text}}(x)$, or their concatenation, with an additional <SPE> token inserted at the beginning. The encoded representation for targets y is constructed in the same way. Please refer to Appendix B for detailed input and prediction formats of the unified decoder.

To obtain general task modeling capability, Uni-Perceiver v2 conducts multi-task learning on various uni-modal and multi-modal tasks. Denoting a set of K tasks as $\{\mathcal{X}_k, \mathcal{Y}_k\}_{k=1}^K$, where $\mathcal{X}_k$ and $\mathcal{Y}_k$ are the input set and target set of the k -th task, respectively. The training loss is

$$L = \sum_{k=1}^K s_k \mathbb{E}_{\{x, y\} \in \{\mathcal{X}_k, \mathcal{Y}_k\}} \left[-w_k \log \frac{P(x, y)}{\sum_{z \in \mathcal{Y}_k} P(x, z)} \right], \quad (7)$$

where s_k and w_k denote the sampling ratio and loss weight of the k -th task, respectively. The sampling ratio are normalized as $\sum_k s_k = 1$. We refer to Sec. 4.4 for detailed discussions of the sampling strategy. To mitigate the task interference in multi-task training, we follow Uni-Perceiver-MoE [49] to employ the Conditional MoEs with attribute-level routing strategy for effective multi-task training.

Tasks with Localization. Uni-Perceiver v2 can perform localization tasks such as object detection and instance segmentation by decoding the regional representations. Specifically, for each region proposal q_j^{proposal} , its outputted feature from the unified decoder $g(\cdot)$ will be compared with class embeddings to obtain the class prediction as in Eq. (5). The corresponding bounding box q_j^{box} and segmentation mask q_j^{mask} will serve as the localization predictions.

Tasks without Localization. Uni-Perceiver v2 can also handle tasks that do need localization predictions, *e.g.*, image classification, image captioning, image-text retrieval. It follows a similar formulation of Uni-Perceiver for these tasks with two major differences: (1) More expressive and flexible localization clues for images, better facilitating these tasks; (2) Both the image and text encoders can leverage off-the-shelf modality-specific pre-trained models, leading to better performance.

4.4. Sampling Strategy and Improved Optimization

Optimizing generalist models follows the paradigm of multi-task learning, which performs joint training on data from different tasks. Current methods usually mix all tasks in one training iteration [23, 39, 50]. Such *mixed sampling strategy* limits the batch-size of each task, which can be detrimental for tasks that benefit from large batch-size training (*e.g.*, image-text retrieval).

A straightforward solution is to sample only one task per iteration, which we refer as *unmixed sampling strategy*. It can achieve the largest training batch-size. However, when different iterations sample different tasks, the gradients would vary greatly due to the differences in data and tasks, which may bring potential instability to multi-task learning and performance deterioration.

To mitigate the instability issue of unmixed sampling strategy, we propose Task-Balanced Gradient Normalization. The core idea is to balance the gradient of each task, by normalizing the gradient of each iteration and compensating it according to the task sampling ratio.

Suppose the k -th task is sampled at timestep t , the updating of parameters θ using Task-Balanced Gradient Normalization is obtained by modifying the vanilla AdamW [22] as follows:

$$\begin{cases} \mathbf{g}_t \leftarrow \nabla L_{t,k}(\theta_{t-1}) \\ \mathbf{m}_t = (1 - \beta_1) \mathbf{m}_{t-1} + \beta_1 \mathbf{g}_t \\ \mathbf{n}_t = (1 - \beta_2) \mathbf{n}_{t-1} + \beta_2 \mathbf{g}_t^2 \\ \theta_t = \theta_{t-1} - \alpha \frac{\mathbf{m}_t}{\sqrt{\mathbf{n}_t + \varepsilon}} \end{cases} \Rightarrow \begin{cases} \mathbf{g}_t \leftarrow \omega_k \frac{\nabla L_{t,k}(\theta_{t-1})}{\|\nabla L_{t,k}(\theta_{t-1})\|} \\ \mathbf{m}_t = (1 - \beta_1) \mathbf{m}_{t-1} + \frac{\beta_1}{s_k} \mathbf{g}_t \\ \mathbf{n}_t = (1 - \beta_2) \mathbf{n}_{t-1} + \frac{\beta_2}{s_k} \mathbf{g}_t^2 \\ \theta_t = \theta_{t-1} - \alpha \frac{\mathbf{m}_t}{\sqrt{\mathbf{n}_t + \varepsilon}} \end{cases}$$

where $L_{t,k}$ is the loss function for the sampled k -th task at timestep t , and α is the learning rate. The weight decay and bias corrections are omitted for simplicity. The original task gradients are first normalized to stabilize training. The scaling factor ω_k serves as the balance coefficient of the sampled task. Then the trimmed gradient $\mathbf{g}_t$ can be used to estimate the first moment $\mathbf{m}_t$ and second moment $\mathbf{n}_t$ of gradients in a moving average way. To further decouple the gradient contribution and sampling ratio s_k of each task, a task-specific compensation coefficient $1/s_k$ is used to unbiased the estimation $\mathbf{m}_t$ and $\mathbf{n}_t$. In practice, if all tasks are expected to contribute equally, all scaling factors could be set as $\omega_k = 1$.

5. Experiments

5.1. Datasets

Uni-Perceiver v2 performs multi-task training on various tasks and public-available datasets to achieve the general task modeling capability. It uses similar datasets as in Uni-Perceiver [50]. Specifically, the image classification task is trained on ImageNet-1k [8] dataset. For objection detection and instance segmentation, COCO [19] is used for training. For image captioning and image-text retrieval, we use a combination of image-text-pair datasets: SBU Captions [25], Visual Genome [15], COCO Caption [7], CC3M [33], CC12M [4] and YFCC [13]. We also add the language modeling task during training, which is trained on BookCorpus [51] and English Wikipedia (Books&Wiki).

During the evaluation, we evaluate generalist models on the most representative datasets for the pillar vision and vision-language tasks listed in Tab. 1. Specifically, ImageNet-1k [8] and COCO Caption [7] are utilized to evaluate the performance of image classification and image caption, respectively. For image-text retrieval, COCO Caption and Flickr30k [26] are utilized. Note that Flickr30k is not involved in training. For objection detection and instance segmentation, COCO [19] is used to evaluate their performances. We put the licenses of all datasets in the Appendix.

5.2. Implementation Details

We implement three Uni-Perceiver v2 variants with different backbones, *i.e.*, ResNet-50 [11], Swin-Base [21], and Swin-Large. ResNet-50 is pre-trained on ImageNet-1k, and Swin-Base is pre-trained on ImageNet-21k. Swin-Large is firstly pre-trained on ImageNet-21k and then trained on the detection task with Object365 [32]. A Transformer [36]-based region proposal network [16] is used to generate general region proposals. However, we replace all multi-category classifiers with binary classifiers. We choose the pre-trained RoBERTa_{BASE} [20] as the text encoder, which is jointly tuned with the whole network. The unified decoder is also a Transformer-based network, whose parameters are initialized randomly and optimized from scratch. Its architecture follows the setting of the BERT_{BASE} [9] model, but it only consists of 6 Transformer layers. To mitigate the task interference issue in multi-task learning, we also employ the attribute-level Conditional MoE [49] in all FFN layers of the unified decoder.

Unless specifically stated, we adopt the unmixed sampling strategy, which only samples one task for all GPUs per iteration. Please refer to the Appendix B for more training settings and implementation details.

5.3. Ablation Studies

In the following, we evaluate the key components of Uni-Perceiver v2 with ResNet-50 backbone by evaluating its

Representation Types	COCO Detection	ImageNet-1k Classification	COCO Retrieval	COCO Caption
Global	-	76.8	46.3 34.6	28.8
Regional	48.2	75.9	52.3 39.2	31.2
Global + Regional	49.9	76.9	51.3 38.8	30.6

Table 2. Ablation of different representation types for general region proposals. Results are reported on object detection (mAP), image classification (Acc), image-text retrieval (I2T R@1 and T2I R@1), and image caption (BLEU-4).

performance on four tasks, *i.e.*, image detection on COCO, image classification on ImageNet-1k, image-text retrieval on COCO caption, and image captioning on COCO caption. The instance segmentation and language modeling tasks are not included to save training costs, and the YFCC dataset is also excluded from the training. Note that, the performance on these datasets are reported without any task-specific fine-tuning. If not stated, COCO detection pre-trained ResNet-50 is used for ablation studies to accelerate convergence.

Effectiveness of Global and Regional Image Representations. Uni-Perceiver v2 encodes images as the concatenation of global and regional representations. To evaluate their effectiveness on different tasks, we conduct experiments that employ different representations, *i.e.*, only using global representations, only using regional representation only, and using both. Results in Tab. 2 show that: (1) regional representation is crucial for both captioning and retrieval tasks. We speculate that this is because regional proposals can provide localization clues, which is helpful to process both tasks. (2) Compared with regional-only representations, global representations deliver better results on the image classification task, which indicates global representations are important for image-level tasks. (3) Combining global and regional representation allows the two representations to complement each other, and thus achieve the best overall results on all tasks. Therefore, in our subsequent experiments, combining global and regional representations is taken as the default setting.

Task Collaboration and Interference. To analyze the collaboration and interference between different tasks, we conduct experiments by removing each task independently from the joint-training tasks in Tab. 3. If the removal of one task can improve (or degrade) the performance of another task, it can reflect that the former task is detrimental (or beneficial) to the latter one during joint training. For a fair comparison, the Conditional MoEs are not employed except for the last experiment. Results show that without MoEs, other tasks have negative impacts on the training of image-text retrieval. However, the image-text retrieval task could promote the performance of image captioning. The image classification task is also very helpful to image captioning, yet the reverse has no obvious effect. It should be noted that

Tasks	COCO Detection	ImageNet-1k Classification	COCO Retrieval		COCO Caption
Single Task	50.1	76.1	50.0	37.6	30.2
All Tasks	49.8	76.3	46.0	34.7	28.9
w/o Detection	-	76.6 (+0.3)	47.0 (+1.0)	34.6 (-0.1)	30.4 (+0.5)
w/o Classification	50.1 (+0.3)	-	51.6 (+5.6)	38.6 (+3.9)	25.9 (-3.0)
w/o Retrieval	49.5 (-0.3)	76.3 (+0.0)	-	-	27.4 (-1.5)
w/o Captioning	49.7 (-0.1)	76.3 (+0.0)	51.2 (+5.2)	38.3 (+3.6)	-
All Tasks w/ MoE	49.9 (+0.1)	76.9 (+0.6)	51.3 (+5.3)	38.8 (+4.1)	30.6 (+0.7)

Table 3. Ablation of collaboration and interference between tasks. All experiments except for the last line do not employ Conditional MoEs. In the brackets are the gaps to the “All Tasks” counterpart. In **green** and **red** are the gaps of at least ± 0.5 point.

Task	Gather	TBGN	COCO Detection	ImageNet-1k Classification	COCO Retrieval	COCO Caption
Sampling	Feature					
mixed			49.6	76.7	40.1 31.9	27.6
unmixed			49.2	76.6	39.8 30.9	27.5
unmixed	✓		49.3	76.8	50.4 37.3	27.6
unmixed	✓	✓	49.9	76.9	51.3 38.8	30.6

Table 4. Ablation of sampling strategies and improved optimizer. “mixed” means mixing different tasks’ data in one iteration, while “unmixed” denotes that only one task’s data is sampled in one iteration. “Gather Feature” means that negative samples for retrieval tasks are collected synchronously across GPUs. “TBGN” denotes Task-Balanced Gradient Normalization.

all models employ an image encoder pre-trained on COCO detection, thereby all these tasks can benefit from the pre-trained region proposal network. The results indicate that task interference indeed exists in the multi-task training of generalist models and is more common than task collaboration, suggesting the importance of addressing the task interference issue. By employing Conditional MoEs, the task interference is largely mitigated, resulting in improved results on all tasks.

Sampling Strategy and Improved Optimization. We evaluate the effectiveness of the unmixed sampling strategy (*i.e.*, sampling one task for each iteration) and the proposed Task-Balanced Gradient Normalization in Tab. 4. From the results, we observe that the vanilla unmixed sampling strategy that computing the contrastive loss with samples on each GPU have slightly adverse effect on the learning of all tasks when compared with the mixed sampling strategy. With the batch size increased by gathering features across all GPUs, the performance of retrieval tasks can be largely improved. Further introducing Task-Balanced Gradient Normalization leads to more stable multi-task training and consistently improved performance across all tasks.

Effects of Different Image Encoder Pre-training. By integrating off-the-shelf encoder models, Uni-Perceiver v2 is capable of leveraging existing large-scale pre-trained encoders. To analyze the effects of different pre-training,

Pretrained Method	Pretrained Data	COCO Detection	ImageNet-1k Classification	COCO Retrieval	COCO Caption
Supervised	IN-1k	45.7	76.8	51.2 38.9	27.3
Supervised	IN-21k	48.3	80.1	55.1 41.2	30.2
Supervised	IN-1k & COCO	49.9	76.9	51.3 38.8	30.6
MoCo v2	IN-1k	48.3	75.0	54.8 40.5	29.6
CLIP	CLIP data	47.2	73.8	55.3 41.3	32.0

Table 5. Ablation of different pre-trained image encoders.

we employ different pre-trained models for image encoders. For models with supervised pre-training, we employ ResNet-50 pre-trained on ImageNet-1k, on ImageNet-21k, or consecutively pre-trained on ImageNet-1k and COCO. For models with weakly-supervised or unsupervised pre-training, we employ ResNet-50 pre-trained with MoCo v2 [6] or CLIP [27]. Tab. 5 demonstrates that different pre-training data and methods of image encoders benefit different downstream tasks. Specifically, supervised pre-training methods show the most obvious benefits on downstream tasks similar to it, *e.g.*, ImageNet-21k pre-training delivers the best results on ImageNet-1k classification. Besides, the pre-training on large-scale supervised (ImageNet-21k), weakly-supervised or unsupervised data (CLIP and MoCo v2) is more helpful to vision-language tasks such as image-text retrieval and image captioning, which possibly thanks to more general representations.

5.4. Main Results

To further verify the effectiveness of Uni-Perceiver v2, we incorporate more powerful backbones including Swin-Base and Swin-Large, denoted as Uni-Perceiver-v2_{BASE} and Uni-Perceiver-v2_{LARGE}, respectively. In addition to the tasks included in the ablation studies, we also incorporate instance segmentation on COCO, language modeling on Books&Wiki, and image captioning / image-text retrieval on YFCC for larger-scale multi-task training.

Comparison with existing Generalist Models. We list the performance of Uni-Perceiver v2 and other generalist models on pillar vision and vision-language tasks in Tab. 6. Since generalist models aim to process different tasks with shared architecture and parameters, the task-specific fine-tuning will lose the general modeling ability. We report the performance of the shared models without any task-specific adaptation. Specifically, Uni-Perceiver-v2_{BASE} can outperform all previous generalist models on all tasks except the Flickr30k retrieval, even if some methods have $> 10\times$ model parameters, *e.g.*, Unified-IO_{XL} and Flamingo-3B. The performance disadvantage on Flickr30k may be due to the use of private data by Flamingo-3B. Further Scaling up to Swin-Large backbone, Uni-Perceiver-v2_{LARGE} obtains the best performance on all tasks. Thanks to the flexibility of general region proposals, Uni-Perceiver v2 supports most pillar tasks among generalist models and can achieve

Methods	#params	Image	Object	Instance	Image	Text		Image	
		Classification	Detection	Segmentation	Captioning	Retrieval		Retrieval	
		ImageNet-1k Acc	COCO mAP	COCO mAP	COCO B@4 CIDEr	COCO R@1	Flickr30k R@1	COCO R@1	Flickr30k R@1
Pix2Seq v2 [5]	132M	-	<u>46.5</u>	<u>38.2</u>	34.9 -	-	-	-	-
UniTab [43]	185M	-	-	-	- 115.8	-	-	-	-
Unified-IO _{LARGE} [23]	776M	71.8	-	-	- -	-	-	-	-
Unified-IO _{XL} [23]	2.9B	79.1	-	-	<u>122.3</u>	-	-	-	-
Flamingo-3B [1]	3.2B	-	-	-	- -	65.9	<u>89.3</u>	48.0	<u>79.5</u>
Uni-Perceiver _{BASE} [50]	124M	79.2	-	-	32.0 -	64.9	82.3	50.7	71.1
Uni-Perceiver _{LARGE} [50]	354M	82.7	-	-	35.3 -	67.8	83.7	54.1	74.2
Uni-Perceiver-MoE _{BASE} [49]	167M	80.3	-	-	33.2 -	64.6	82.1	51.6	72.4
Uni-Perceiver-MoE _{LARGE} [49]	505M	<u>83.4</u>	-	-	<u>35.5</u> -	<u>67.9</u>	83.6	<u>55.3</u>	75.9
Uni-Perceiver-v2 _{BASE}	308M	86.3	58.6	50.6	35.4 116.9	71.8	88.1	55.6	73.8
Uni-Perceiver-v2 _{LARGE}	446M	87.2 (+3.8)	61.9 (+15.4)	53.6 (+15.4)	36.5 122.5 (+1.6) (+0.2)	75.0 (+7.1)	89.3 (+0.0)	58.5 (+3.2)	79.6 (+0.1)

Table 6. Comparison of our Uni-Perceiver v2 to recent generalist models on six pillar visual and visual-linguistic tasks listed in Tab. 1. Note that we only report the results without any task-specific fine-tuning. Uni-Perceiver v2 is the the first generalist model to support all these pillar tasks and can achieve competitive results without any task-specific adaption. Some generalist models that only report results with task-specific fine-tuning are not included, *e.g.*, OFA [39] and GIT [38]. “#params” is the number of parameters required during model deployment for cross-modal tasks. Results with the best performance are in **bold**, and previous SoTA results are underlined.

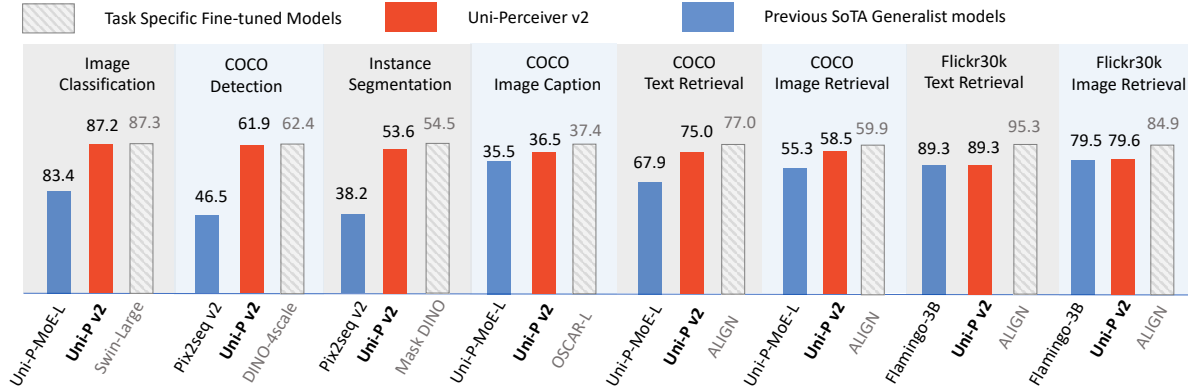


Figure 2. Comparison with generalist models and commonly-recognized strong task-specific models on pillar vision and vision-language tasks. For generalist models including Uni-Perceiver v2, we only report the results without any task-specific fine-tuning. Uni-Perceiver v2 (Uni-P v2) is compared with competitive specialized models, *i.e.*, Swin-Large [21], DINO [47], Mask DINO [16], OSCAR-L [18] and ALIGN [12], and previous SoTA generalists, *i.e.*, Uni-P-MoE-L [49], Pix2seq v2 [5], and Flamingo-3B [1].

competitive results consistently, which indicates the superior general modeling performance of Uni-Perceiver v2 in both versatility and performance.

Comparison with Specialized Models. We compare Uni-Perceiver v2 with commonly-recognized strong baseline models and previous SoTA generalist models on the pillar tasks in Tab. 2. The results show that Uni-Perceiver v2 significantly decreases the performance gap between generalist models and commonly-recognized strong baselines, which need task-specific fine-tuning. It can achieve comparable results across all tasks except the retrieval task on Flickr30K, which we suspect is because ALIGN [12] uses 1.8B private image-text pairs for training, which is much larger than ours that uses only publicly available datasets.

6. Conclusion

We propose Uni-Perceiver v2, which is the first generalist model that achieves competitive results on major large-scale vision and vision-language tasks. After being jointly trained on single-modal and multi-modal tasks, Uni-Perceiver v2 achieves competitive performance on a broad range of downstream tasks. As for **limitations**, our method has not been verified on image generation tasks due to limited computational resources.

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